Transfer of Reed-Muller local testing to multiplicity codes with quantitative soundness

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Editorial scope. Counted original Theorem51 (introduced as Theorem4): quantitative transfer from an assumed RM k-dimensional tester to the MRM tester, with its exact alpha-prime coefficient and parameter restrictions. Full Claims53–56 and final derivative reconstruction are included. No support claim or numerical corollary is counted separately. Original author mathematics is retained as published.

Allowed standard prerequisites (editorial).

- Reed-Muller distance bound [10], standard Hasse derivative restriction identity [15], Markov and Schwartz-Zippel bounds. Author Lemmas11,15,17 and exact alphabet-reduction definition48 retained. Main statement retains the assumed RM tester soundness instead of claiming it unconditionally.

2 Preliminaries (author context excerpts)

2.1 Notation

We denote vectors by bold letters. For $\boldsymbol{X} = X_1, \ldots, X_m$ we denote by $\mathbb{F}[\boldsymbol{X}]$ the set of multivariate polynomials in the variables $X_1, \ldots, X_m$. We denote by $\mathbb{F}[\boldsymbol{X}]^{\leq d}$ the set of polynomials of individual degree at most d, and by $\mathbb{F}[\boldsymbol{X}]^{\mathrm{loc}\leq d}$ the set of polynomials of individual degree at most d (i.e degree in each variable). Given a vector $\boldsymbol{I} = (i_1, \ldots, i_m) \in \mathbb{N}^m$, we use the notation

$$\boldsymbol{X}^{\boldsymbol{I}} \stackrel{\mathrm{def}}{=} \prod_{k=1}^{m} X_k^{i_k}.$$

2.2 Reed Muller and Multiplicity codes

Definition 10. Let d, m be non-negative integers, and q a prime power. The (m, d, q) Reed-Muller code is defined as the set of evaluation vectors of m-variate polynomials of degree $\leq d$ over $\mathbb{F}_q^m$, namely,

$$\mathrm{RM}(q, m, d) = \{(f(\alpha))_{\alpha \in \mathbb{F}_q^m} \mid f \in \mathbb{F}_q[\boldsymbol{X}]^{\leq d}\}.$$

We will make use of the following lemma:

Lemma 11 ([10], Lemma 3.2). Let $\delta^{\mathrm{RM}}_{q,m,d}$ be the relative distance of the code $\mathrm{RM}(q,m,d)$. Then $\delta^{\mathrm{RM}}_{q,m,d} \geq q^{-d/(q-1)}$.

Definition 12 (Hasse derivative). For a multivariate $P(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ where $\boldsymbol{X} = (X_1, \ldots, X_m)$ for some $m \in \mathbb{N}$, and a non-negative vector $\boldsymbol{I} \in \mathbb{N}^m$, the $\boldsymbol{I}$-th Hasse derivative of P, denoted $P^{(\boldsymbol{I})}(\boldsymbol{X})$, is the coefficient of $\boldsymbol{Z}^{\boldsymbol{I}}$ in the polynomial $P(\boldsymbol{X}, \boldsymbol{Z}) = P(\boldsymbol{X} + \boldsymbol{Z})$. Thus

$$P(\boldsymbol{X} + \boldsymbol{Z}) = \sum_{\boldsymbol{I}} P^{(\boldsymbol{I})}(\boldsymbol{X}) \cdot \boldsymbol{Z}^{\boldsymbol{i}}.$$

Hasse derivatives are linear. I.e, for all $P, Q \in \mathbb{F}[\boldsymbol{X}]$ and $\lambda \in \mathbb{F}$, $(\lambda P)^{(\boldsymbol{I})}(\boldsymbol{X}) = \lambda P(\boldsymbol{X})$ and $P^{(\boldsymbol{I})}(\boldsymbol{X}) + Q^{(\boldsymbol{I})}(\boldsymbol{X}) = (P+Q)^{(\boldsymbol{I})}(\boldsymbol{X})$. The product rule shows $(PQ)^{(\boldsymbol{I})}(\boldsymbol{X}) = \sum_{\boldsymbol{I}_0 + \boldsymbol{I}_1 = \boldsymbol{I}} P^{(\boldsymbol{I}_0)}(\boldsymbol{X}) \cdot Q^{\boldsymbol{I}_1}(\boldsymbol{X})$.

Definition 13 (Weight). If $\boldsymbol{I} = (i_1, \ldots, i_m) \in \mathbb{N}^m$ then $w(\boldsymbol{I}) = \sum_{j=1}^m i_j$.

Definition 14 (Multiplicity). For $P(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ and $\boldsymbol{a} \in \mathbb{F}^m$, the multiplicity of P at $\boldsymbol{a}$, denoted $\mathrm{mult}(P, \boldsymbol{a})$, is the largest integer s such that for every non-negative vector $\boldsymbol{I}$ with $w(\boldsymbol{I}) < s$ we have $P^{(\boldsymbol{I})}(\boldsymbol{a}) = 0$. If s may be taken arbitrarily large, we set $\mathrm{mult}(P, \boldsymbol{a}) = \infty$. Note that by definition $\mathrm{mult}(P, \boldsymbol{a}) \geq 0$ for every $\boldsymbol{a}$. One important property about multiplicities is a generalization of the Schwartz-Zippel lemma for multivariate polynomials:

Lemma 15 ([6]). Let $P \in \mathbb{F}[\boldsymbol{X}]$ be a non-zero polynomial of total degree at most d. Then for any finite $A \subseteq \mathbb{F}$,

$$\sum_{\boldsymbol{a} \in A^m} \mathrm{mult}(P, \boldsymbol{a}) \leq d \cdot |A|^{m-1}.$$

Definition 16 (Multiplicity code). Let $m, d \geq s$ be non-negative integers, and let q be a prime power. Let

$$\Sigma_{m,s} = \mathbb{F}_q^{\{\boldsymbol{I} : w(\boldsymbol{I}) < s\}} \simeq \mathbb{F}_q^{\binom{m+s-1}{m}}.$$

For $P(\boldsymbol{X}) \in \mathbb{F}_q[X_1, \ldots, X_m]$ we define the order s evaluation of P at $\boldsymbol{a}$, denoted $P^{(<s)}(\boldsymbol{a})$, to be the vector $(P^{(\boldsymbol{I})}(\boldsymbol{a}))_{\boldsymbol{I} : w(\boldsymbol{I}) < s} \in \Sigma_{m,s}$. The multiplicity code $\mathrm{MRM}(q, m, d, s)$ is defined as follows. The alphabet of the code is $\Sigma_{m,s}$ and the length is q^m. Every polynomial $P(\boldsymbol{X}) \in \mathbb{F}_q[\boldsymbol{X}]$ of $\deg(P) \leq d$ defines a codeword by $(P^{(<s)}(\boldsymbol{a}))_{\boldsymbol{a} : \boldsymbol{a} \in \mathbb{F}_q^m} \in (\Sigma_{m,s})^{q^m}$. We also let $\mathrm{MRS}(q, d, s) := \mathrm{Mult}(q, 1, d, s)$ stand for Reed-Solomon multiplicity code. The following lemma states the relationship between the derivatives of a polynomial to the derivatives of its restriction to a line.

Lemma 17 ([15], Sec 4). Let $P \in \mathbb{F}[\boldsymbol{X}]$ be a multivariate polynomial where $\boldsymbol{X} = (X_1, \ldots, X_m)$. Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{F}^m$ and define a univariate polynomial $P|_{L_{\boldsymbol{a},\boldsymbol{b}}}(t) = P(\boldsymbol{a}t + \boldsymbol{b})$. Then

$$(P|_{L_{\boldsymbol{a},\boldsymbol{b}}})^{(j)}(t) = \sum_{\boldsymbol{I} : w(\boldsymbol{I}) = j} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}) \cdot \boldsymbol{a}^{\boldsymbol{I}}.$$

3 Polynomials and tables (author context excerpts)

A table is an element of $\Sigma_{m,s}^{q^m}$, i.e., a function mapping every evaluation point in $\mathbb{F}_q^m$ to an element in $\Sigma_{m,s}$. EVAL takes a multi-variate polynomial and returns its table of evaluations. More precisely,

Definition 39. We define $\mathrm{EVAL}_{m,s} : \mathbb{F}_q[\boldsymbol{X}] \times \mathbb{F}_q^m \to \Sigma_{m,s}$ by

$$\mathrm{EVAL}_{m,s}(P; \boldsymbol{a}) = (P^{(\boldsymbol{I})}(\boldsymbol{a}))_{w(\boldsymbol{I})<s}.$$

Similarly, we define $\mathrm{EVAL}_{m,s} : \mathbb{F}_q[\boldsymbol{X}] \to (\Sigma_{m,s})^{q^m}$ by

$$\mathrm{EVAL}_{m,s}(P) = (\mathrm{EVAL}(P; \boldsymbol{a}))_{\boldsymbol{a}\in\mathbb{F}_q^m}.$$

We say $T \in (\Sigma_{m,s})^{q^m}$ is the table of $P \in \mathbb{F}_q[\boldsymbol{X}]$ if $\mathrm{EVAL}(P) = T$. An element in $\Sigma_{m,s}$ contains information about all the derivatives of order up to s. Sometimes we would like to focus on a certain directional derivative. If $\sigma \in \Sigma_{m,s}$ and $\boldsymbol{I} \in \mathbb{N}^m$ with $w(\boldsymbol{I}) < s$, then $\sigma_{(\boldsymbol{I})} \in \mathbb{F}_q$ is the entry of σ that encodes the $\boldsymbol{I}$'th derivative. Similarly, if $f \in \Sigma_{m,s}^{q^m}$ is a table, i.e., $f : \mathbb{F}_q^m \to \Sigma_{m,s}$, then $f_{(\boldsymbol{I})} : \mathbb{F}_q^m \to \mathbb{F}_q$ is defined by letting $f_{(\boldsymbol{I})}(\boldsymbol{x})$ be the $\boldsymbol{I}$'th entry of $f(\boldsymbol{x}) \in \Sigma_{m,s}$. To understand the low-dimensional tests on tables, we need to define the restriction of tables to subspaces. If T is a table $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ and we want to restrict to the plane $\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}$ the restriction $T_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}$ should be a table $\mathbb{F}_q^2 \to \Sigma_{2,s}$. To this end we define the alphabet reduction map $\phi_{(\boldsymbol{a},\boldsymbol{b})} : \Sigma_{m,s} \to \Sigma_{2,s}$ by

Definition 48.

$$(\phi_{(\boldsymbol{a},\boldsymbol{b})})(\boldsymbol{z})_{\boldsymbol{J}=(j_1,j_2)} = \sum_{\boldsymbol{I}\in\mathbb{N}^m} z_{\boldsymbol{I}} \cdot \sum_{\substack{\boldsymbol{I}_1+\boldsymbol{I}_2=\boldsymbol{I} \\ w(\boldsymbol{I}_1)=j_1, w(\boldsymbol{I}_2)=j_2}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1} \boldsymbol{a}^{\boldsymbol{I}_1}\boldsymbol{b}^{\boldsymbol{I}_2}.$$

This map applies the "chain rule" to an element in $\Sigma_{m,s}$, in accordance with Lemma 18. We may then define

$$T_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} = \phi_{(\boldsymbol{a},\boldsymbol{b})} \circ T \circ \ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}.$$

Similarly, for $\boldsymbol{h} = (\boldsymbol{h}_0, \boldsymbol{h}_1, \ldots, \boldsymbol{h}_k)$ we define $\phi_{\boldsymbol{h}}$ and $T_{\boldsymbol{h}}$ by:

$$(\phi_{\boldsymbol{h}}(\boldsymbol{z}))_{\boldsymbol{J}} = \sum_{\boldsymbol{I}\in\mathbb{N}^m} z_{\boldsymbol{I}} \cdot \sum_{\substack{\boldsymbol{I}_1+\cdots+\boldsymbol{I}_k=\boldsymbol{I} \\ w(\boldsymbol{I}_r)=j_r}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1, \ldots, \boldsymbol{I}_k} \prod_{i=1}^{k} \boldsymbol{h}_k^{\boldsymbol{I}_k}, \text{ and,}$$

$$T_{\boldsymbol{h}}(\boldsymbol{Y}) = \phi_{(\boldsymbol{h})} \circ T\left(\boldsymbol{h}_0 + \sum_{i=1}^{k} Y_i \boldsymbol{h}_i\right).$$

5 Local testing above the RM testing dimension

In this section we look at the local characterization and local testing of $\mathrm{MRM}(q, m, d, s)$ by dimension k tests, when k is above the RM testing dimension $t_{q,d} = \lceil (d+1)/(q - q/p) \rceil$ of $\mathrm{RM}(q, m, d)$. By Theorem 2 dimension k subspaces give a local

test (and hence also a local characterization) for $\mathrm{RM}(q,m,d)$. We show they also give a local test (and hence also a local characterization) for $\mathrm{MRM}(q,m,d,s)$ for $s<q$. There is, however, some parameter loss in the reduction as we next explain. To formally state the result we need some notation. For $\boldsymbol{x}\in\mathbb{F}_q^m$ let

$$H_k=\{\boldsymbol{h}=(\boldsymbol{h}_0,\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k)\mid \boldsymbol{h}_0,\ldots,\boldsymbol{h}_k\in\mathbb{F}_q^m,\dim(\mathrm{span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\})=k\},\text{ and,}$$
$$H_{k,\boldsymbol{x}}=\{\boldsymbol{h}\in H_k\mid \boldsymbol{x}\in\boldsymbol{h}_0+\mathrm{Span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\}\}.$$

By $\boldsymbol{h}\sim_k H$ (resp. $H_{k,\boldsymbol{x}}$) we mean a choosing $\boldsymbol{h}$ uniformly at random from H_k (resp. $H_{k,\boldsymbol{x}}$). We let $A_{\boldsymbol{h}}=\boldsymbol{h}_0+\mathrm{Span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\}$ be the k-dimensional affine space defined by $\boldsymbol{h}$. We also recall the distance and rejection function from the introduction:

Definition 50. Let $T:\mathbb{F}_q^m\to\Sigma_{m,s}$ be a table. Then

- $\delta(T,\mathrm{MRM}(q,m,d,s))$ is the distance between T and the closest evaluation table of a degree d polynomial, i.e., $\delta(T,\mathrm{MRM}(q,m,d,s))=\min_{G\in\mathrm{MRM}(q,m,d,s)}\{\delta(T,G)\}$, and,
- $\mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)$ is the probability a dimension-k test demonstrates that T is not a degree d polynomial.

With this notation we prove:

Theorem 51. Let $\mathbb{F}_q$ be a field of size q, and assume $s\le\min\{d,q-1\}$. Suppose for $\mathrm{RM}(q,m,d)$ there exists $\alpha>0$ and $c_0\le 1$ such that for every f

$$\mathrm{REJ}_{k,d}^{\mathrm{RM}}(f)\ge\min\{\alpha\cdot\delta(f,\mathrm{RM}(q,m,d)),c_0\}.$$

Then, for every $s<q$, for every T we have

$$\mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)\ge\min\{\alpha'\cdot\delta(T,\mathrm{MRM}(q,m,d,s)),c_0\} \tag{6}$$

for

$$\alpha'=\left(1-\frac{s-1}{q}\right)\frac{1}{1+q^{d/(q-1)\frac{1}{\alpha}}}. \tag{7}$$

For example, assume q is prime. If $d+1\le q-1$ the testing dimension is 1, meaning that lines are a good local test for $\mathrm{RM}(q,m,d)$. Then, Theorem 51 says that lines are also a good test for $\mathrm{MRM}(q,m,d,s)$, alas, with a larger coefficient. Similarly, if $q<d\le 2(q-1)$ the RM testing dimension is 2 and therefore planes are a good local test for $\mathrm{RM}(q,m,d)$. Then, Theorem 51 says that planes are also a good test for $\mathrm{MRM}(q,m,d,s)$, with a larger coefficient. In the same vein 3-dimensional planes are a good local test for $\mathrm{MRM}(q,m,d,s)$ when $d\le 3(q-1)$. To explain the intuition behind the theorem we first consider the characterization aspect of it. Suppose k is above the RM testing dimension, and that the table T passes all $\mathrm{MRM}(q,m,d,s)$ k-dimensional tests $\boldsymbol{h}$. Specifically, this means that for every $\boldsymbol{h}\in H_k$, the table T restricted to the k-dimensional affine space $A_{\boldsymbol{h}}$ is consistent with a degree d polynomial $P_{\boldsymbol{h}}$. Now, let $T_{(0)}$ be the table T where at each entry we keep only the evaluation of the function itself and remove the evaluations that are associated with higher derivatives. Then, in particular, for every $\boldsymbol{h}\in H_k$, the table $T_{(0)}$ restricted to the k-dimensional

space $A_{\boldsymbol{h}}$ is still consistent with the degree d polynomial $P_{\boldsymbol{h}}$. As k is above the RM testing dimension, there must be a unique degree d polynomial P that is consistent with $P_{\boldsymbol{h}}$ (and the table $T_{(0)}$) for every $\boldsymbol{h} \in H_k$. This P is the only possible candidate for a low-degree explanation of the table T. What we need to check, and is indeed correct, is that since T passes all dimension k tests, P is indeed consistent with T. The testing case requires more technical details but is similar in spirit. We first, again, look at $T_{(0)}$ that contains only the function evaluations, and not the higher derivatives evaluations. If T passes the test with high enough probability, then so does $T_{(0)}$, and this ensures the existence of a global degree-d polynomial P that agrees with most values of $T_{(0)}$. Again, what remains to be shown is that P agrees with most values of T, which we indeed prove. In short, one can informally say that Theorem 51 shows that the MRM testing dimension is not larger than the RM testing dimension, and that above the RM testing dimension one can get local testing for MRM codes. A natural question is whether the MRM testing dimension is equal to the RM testing dimension, or not. In Section 4 we saw that when the RM testing dimension is larger than 1, so is the MRM testing dimension, and lines do not characterize the MRM code. One might be drawn to the conjecture that the RM and MRM testing dimensions coincide. However, surprisingly, in Section 6 we show that no matter what the RM testing dimension is, when $d < sq$ the MRM testing dimension is at most two (for a precise statement see Theorem 60). For example, for $\mathrm{MRM}(q, d, m, s)$ with $2q < d < 3(q-1)$, the RM testing dimension is (roughly) 3 while the MRM testing dimension is still 2.

5.1 The proof

Proof. Let $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ be a table. Let

$$\rho = \mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)$$

be the k-dimensional MRM test rejection probability. If $\rho \geq c_0$ we are done. Therefore, we assume $\rho < c_0$. We first utilize what we know at the zero level.

Claim 53. Let $\delta^{\mathrm{RM}} = \delta_{q,m,d}^{\mathrm{RM}}$ be the distance of the $\mathrm{RM}(q, m, d)$ code. There exists a degree d polynomial P such that

$$\Pr_{\boldsymbol{h} \in H_k}\left((\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} \neq \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})\right) \stackrel{\mathrm{def}}{=} \varepsilon_0 \leq \rho\left(1 + \frac{1}{\alpha \cdot \delta^{\mathrm{RM}}}\right).$$

We call $\boldsymbol{h}$ good if $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$ and bad otherwise. In this terminology, $\varepsilon_0 = \Pr_{\boldsymbol{h} \in H_k}[\boldsymbol{h} \text{ is bad}]$.

Proof. Let $T_{(0)} : \mathbb{F}_q^m \to \mathbb{F}_q$ be as in Section 3, i.e., the table where we keep only the entries of the function evaluations and ignore the evaluations of higher order derivatives. For a given affine space A we have that $(T|_A)_{(0)} = T_{(0)}|_A$, and so if the former agrees with a degree d polynomial, so does the latter. It follows that

$$\mathrm{REJ}_{k,d}^{\mathrm{RM}}(T_{(0)}) \leq \rho.$$

By the hypothesis regarding $\mathrm{RM}(q, m, d)$ and using $\rho < c_0$ there exists a unique degree d polynomial $P \in \mathbb{F}_q[\boldsymbol{X}]$ such that

$$\delta_{(0)} \stackrel{\mathrm{def}}{=} \delta(T_{(0)}, P) \leq \rho/\alpha.$$

We notice that

$$\begin{aligned}\mathbb{E}_{\boldsymbol{h}\sim H_k}\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) &= \Pr_{\boldsymbol{x}\sim\mathbb{F}_q^m}(T_{(0)}(\boldsymbol{x}) \neq P(\boldsymbol{x})) \\ &= \delta(T_{(0)}, P) = \delta_{(0)},\end{aligned}$$

because the subspaces in $A_{\boldsymbol{h}}$ for $\boldsymbol{h} \in H_k$, cover every point an equal number of times. Therefore, by Markov's inequality,

$$\Pr_{\boldsymbol{h}\sim H_k}(\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) \geq \delta^{\mathrm{RM}}) \leq \frac{\delta_{(0)}}{\delta^{\mathrm{RM}}}. \tag{8}$$

Also, by assumption,

$$\Pr_{\boldsymbol{h}\in H_k}((\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} \notin \mathrm{MRM}(q,k,d,s)) \leq \rho. \tag{9}$$

This means that except for probability ρ over $\boldsymbol{h}$, $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}}$ agrees with $\mathrm{EVAL}(P_{\boldsymbol{h}})$ for some k-variate degree d polynomial $P_{\boldsymbol{h}}$. When this happens it also holds that $T_{(0)}$ agrees with $P_{\boldsymbol{h}}$ over $A_{\boldsymbol{h}}$ (because $\phi_{\boldsymbol{h}}$ does not change the zero level). Thus, Equations (8) and (9) together imply that except for probability $\rho + \delta_0/\delta^{\mathrm{RM}} \leq \rho(1 + 1/(\alpha\delta^{\mathrm{RM}}))$ over $\boldsymbol{h}$, we simultaneously have that $T_{(0)} = P_{\boldsymbol{h}}$ over $A_{\boldsymbol{h}}$ and that $\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) < \delta^{\mathrm{RM}}$. When both events happen we conclude that

$$\delta(P_{\boldsymbol{h}}, P|_{A_{\boldsymbol{h}}}) < \delta^{\mathrm{RM}}.$$

As $P_{\boldsymbol{h}}$ and $P|_{A_{\boldsymbol{h}}}$ are degree d polynomials on $A_{\boldsymbol{h}}$ and are closer than δ^{RM}, it must be the that in fact $P_{\boldsymbol{h}} = P|_{A_{\boldsymbol{h}}}$. Thus, $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}}$ is a valid $\mathrm{MRM}(q,k,d,s)$ table, and it is the table of the polynomial $P|_{A_{\boldsymbol{h}}}$, i.e., $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$ as desired. $\triangleleft$ Our goal is to bound $\delta(\mathrm{EVAL}(P), T)$. This means that at most $\boldsymbol{x} \in \mathbb{F}_q^m$ we should have $T(\boldsymbol{x}) = \mathrm{EVAL}(P; \boldsymbol{x})$. $T(\boldsymbol{x})$ and $\mathrm{EVAL}(P; \boldsymbol{x})$ contains values for all the m-variate directional derivatives of order up to s. Our handle on these values is Claim 53, that shows that most $\boldsymbol{h}$ are good, meaning that the k-variate derivatives of $P|_{A_{\boldsymbol{h}}}$ and $\phi_{\boldsymbol{h}} \circ T|_{A_{\boldsymbol{h}}}$ are the same. We notice that every such k-variate derivative is a linear combination (dependent on $\boldsymbol{h}$) of the m-variate derivatives. If $\boldsymbol{x}$ is such that for many $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$, $\boldsymbol{h}$ is good, then for that $\boldsymbol{x}$ we get many linear equations on the m-variate derivatives. Our task is to prove that for many $\boldsymbol{x}$ there are enough good $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$ to force the underlying m-variate derivatives of P and T to agree.

Claim 54. We say $\boldsymbol{x} \in \mathbb{F}_q^m$ is bad if $\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}[\boldsymbol{h} \text{ is bad}] \geq 1 - (s-1)/q$ and good otherwise. Then

$$\Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}[\boldsymbol{x} \text{ is bad}] \leq \frac{q}{q-(s-1)} \cdot \Pr_{\boldsymbol{h}\in H_k}[\boldsymbol{h} \text{ is bad}].$$

Proof. We have

$$\mathbb{E}_{\boldsymbol{x}\in\mathbb{F}_q^m}\left[\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad})\right] = \Pr_{\boldsymbol{h}\in H_k}(\boldsymbol{h} \text{ is bad}) = \varepsilon_0,$$

where ε_0 is as in Claim 53. This is because choosing a uniform $\boldsymbol{h} \in H_k$ is the same as first choosing a uniform $\boldsymbol{x} \in \mathbb{F}_q^m$ and then choosing a uniform $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$.

Therefore, for every $c > 1$, by Markov's inequality,

$$\Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}\left[\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad}) \geq c\cdot\varepsilon_0\right] \leq 1/c.$$

Choosing $c = (q-(s-1))/(q\cdot\varepsilon_0)$ gives the result. ◁ We note that if most k-dimensional subspaces are good, then most lines are:

Claim 55. For every $\boldsymbol{x}\in\mathbb{F}_q^m$ and $k \geq 1$,

$$\Pr_{\boldsymbol{u}\in H_{1,\boldsymbol{x}}}[\boldsymbol{u} \text{ is bad}] \leq \Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}[\boldsymbol{h} \text{ is bad}].$$

Proof. Fix $\boldsymbol{x}$. For every $\boldsymbol{h}\in H_{k,\boldsymbol{x}}$, If $\boldsymbol{h}$ is good, then $(\phi_{\boldsymbol{h}}\circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$. This implies, in particular, that for every $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ such that $A_{\boldsymbol{u}}\subseteq A_{\boldsymbol{h}}$, we also have that $(\phi_{\boldsymbol{u}}\circ T)|_{A_{\boldsymbol{u}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}})$. The result then follows because we can sample $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ by first sampling $\boldsymbol{h}\in A_{k,\boldsymbol{x}}$ and then choosing a random $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ such that $A_{\boldsymbol{u}}\subseteq A_{\boldsymbol{h}}$. ◁ We now fix any good $\boldsymbol{x}$. We will prove:

Claim 56. Suppose $\boldsymbol{x}\in\mathbb{F}_q^m$ is good. Then for any m-variate direction $\boldsymbol{I}$ with $w(\boldsymbol{I}) < s$ we have

$$T_{(\boldsymbol{I})}(\boldsymbol{x}) = P^{(\boldsymbol{I})}(\boldsymbol{x}). \tag{10}$$

Once we prove the claim we can conclude the proof of the theorem because:

$$\begin{aligned}\delta(T,\mathrm{EVAL}(P)) &\leq \Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}(\boldsymbol{x} \text{ is bad}) \leq \frac{q}{q-(s-1)}\cdot\varepsilon_0\\ &\leq \frac{q}{q-(s-1)}\cdot\rho\cdot\left(1+\frac{1}{\alpha\cdot\delta^{\mathrm{RM}}}\right)\\ &\leq \rho\left(1-\frac{q}{s-1}\right)^{-1}\left(1+\frac{q^{d/(q-1)}}{\alpha}\right).\end{aligned}$$

Proof of Claim 56. Fix a good $\boldsymbol{x}\in\mathbb{F}_q^m$ and let $w_0 < s$. We will show Equation (10) simultaneously for all m-variate directions $\boldsymbol{I}$ with $w(\boldsymbol{I}) = w_0$. We know that

$$\Pr_{\boldsymbol{u}\in H_{1,\boldsymbol{x}}}(\boldsymbol{u} \text{ is bad}) \leq \Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad}) < 1-\frac{s-1}{q}, \tag{11}$$

where the first inequality is by Claim 55 and the second because $\boldsymbol{x}$ is good. Suppose $\boldsymbol{u} = (\boldsymbol{b},\boldsymbol{a})$ is good (where $\boldsymbol{b},\boldsymbol{a}\in\mathbb{F}_q^m$). Then, by definition, $(\phi_{\boldsymbol{u}}\circ T)|_{A_{\boldsymbol{u}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}})$. In particular $\phi_{\boldsymbol{u}}(T(\boldsymbol{x})) = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}};\boldsymbol{x})$, because $\boldsymbol{x}\in A_\ell$. Now:

- By Definition 48 (and plugging $k=1$)

$$(\phi_{\boldsymbol{u}}(T(\boldsymbol{x})))_{w_0} = \sum_{\boldsymbol{I},w(\boldsymbol{I})=w_0} T(\boldsymbol{x})_{(\boldsymbol{I})}\cdot\boldsymbol{a}^{\boldsymbol{I}},$$

- Also, by Lemma 17 (plugging $t=0$),

$$(\mathrm{EVAL}(P|_{A_{\boldsymbol{u}}};\boldsymbol{x}))_{w_0} = \sum_{\boldsymbol{I},w(\boldsymbol{I})=w_0} P^{\boldsymbol{I}}(\boldsymbol{x})\cdot\boldsymbol{a}^{\boldsymbol{I}}.$$

Thus, a good $\boldsymbol{u} = (\boldsymbol{b}, \boldsymbol{a})$ gives the liner equation

$$\sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (P^{\boldsymbol{I}}(\boldsymbol{x}) - T(\boldsymbol{x})_{(\boldsymbol{I})}) \cdot \boldsymbol{a}^{\boldsymbol{I}} = 0,$$

where the variables are $v_{\boldsymbol{I}} = P^{\boldsymbol{I}}(\boldsymbol{x}) - T(\boldsymbol{x})_{(\boldsymbol{I})}$ for every m-variate direction $\boldsymbol{I}$ of total weight exactly w_0. Thus, Equation (11) implies that

$$\Pr_{\boldsymbol{a} \in \mathbb{F}_q^m \setminus \{\boldsymbol{0}\}} \left(\sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})) \cdot \boldsymbol{a}^{\boldsymbol{I}} = 0 \right) > \frac{s-1}{q}.$$

Now, look at the polynomial $f_{\boldsymbol{x}} \in \mathbb{F}_q[X_1, \ldots, X_m]$ defined by

$$f_{\boldsymbol{x}}(\boldsymbol{a}) = \sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})) \cdot \boldsymbol{a}^{\boldsymbol{I}}.$$

$f_{\boldsymbol{x}}$ is an m-variate, degree w_0 homogeneous polynomial, and it is 0 with probability larger than $(s-1)/q \geq w_0/q = \deg(f_{\boldsymbol{x}})/q$. By the Schwartz-Zippel lemma, it must be the 0 polynomial. Therefore, $T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})$ for all $\boldsymbol{I}$ with $w(\boldsymbol{I}) = w_0$ as desired. ◁

◁

Remark 57. Another way to view the argument, is that each $\boldsymbol{a} \in \mathbb{F}_q^m$ is an evaluation point of a homogeneous $\mathrm{RM}(q, m, w_0)$ codeword, and therefore each good $\boldsymbol{u} = (\boldsymbol{b}, \boldsymbol{a})$ gives a zero coordinate of the codeword. If the number of good $\boldsymbol{u}$ is too large, we get too many zeroes, and therefore the codeword must be the zero codeword, meaning that the values of the variables are zero as we wish.

Remark 58. The argument we use has the information that many k-dimensional restrictions are good, but then chooses to reduce this knowledge to the weaker statement that for many $\boldsymbol{x}$, for many lines passing through $\boldsymbol{x}$, the linear restrictions are good. It seems that using the stronger statement might give a better code and improve the parameters, but we have not succeeded yet in analyzing this.

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